MDS-Assignment2 Report

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1 Yield Classification and Analysis

[ ]:
```
import pandas as pd
df_features = pd.read_csv('secom.data', sep='\s+', header=None)
df_labels = pd.read_csv('secom_labels.data', sep='\s+', header=None,␣
↪names=['result', 'timestamp'])
df = pd.concat([df_features, df_labels['result']], axis=1)
print(df.head())
```

```
         0        1          2          3       4      5         6       7  \
0  3030.93  2564.00  2187.7333  1411.1265  1.3602  100.0   97.6133  0.1242
1  3095.78  2465.14  2230.4222  1463.6606  0.8294  100.0  102.3433  0.1247
2  2932.61  2559.94  2186.4111  1698.0172  1.5102  100.0   95.4878  0.1241
3  2988.72  2479.90  2199.0333   909.7926  1.3204  100.0  104.2367  0.1217
4  3032.24  2502.87  2233.3667  1326.5200  1.5334  100.0  100.3967  0.1235

        8       9  …       581     582     583     584      585     586  \
0  1.5005  0.0162  …       NaN  0.5005  0.0118  0.0035   2.3630     NaN
1  1.4966 -0.0005  …  208.2045  0.5019  0.0223  0.0055   4.4447  0.0096
2  1.4436  0.0041  …   82.8602  0.4958  0.0157  0.0039   3.1745  0.0584
3  1.4882 -0.0124  …   73.8432  0.4990  0.0103  0.0025   2.0544  0.0202
4  1.5031 -0.0031  …       NaN  0.4800  0.4766  0.1045  99.3032  0.0202

      587     588       589  result
0     NaN     NaN       NaN      -1
1  0.0201  0.0060  208.2045      -1
2  0.0484  0.0148   82.8602       1
3  0.0149  0.0044   73.8432      -1
4  0.0149  0.0044   73.8432      -1

[5 rows x 591 columns]
```

[2]:
```
print(df.shape)
print(df['result'].value_counts())
```

```
(1567, 591)
result
-1    1463
```

```
 1     104
Name: count, dtype: int64
```

1.1 Data Science Framework

Method Attempted	R^2 Score	Key Optimization Logic
Lasso Regression	-0.2981	Introduced L_1 regularization for automatic feature selection in an attempt to reduce model complexity.
PCR (PCA + Regression)	0.1130	Compressed highly correlated sensor features into 5 principal components via PCA, effectively filtering out noise.
Lasso + Bias Correction	**0.0157**	Forcibly aligned the predicted mean values of the training and test sets to correct for systematic bias between batches.

1. Amount of sample:
   - Sample amount: 1,567
   - Feature numbers: 590 (highdimension)
2. Data class:
   - Pass (-1): 1,463 (93.4%)
   - Fail (1): 104 (6.6%)
   - Unbalanced dataset!
3. Data missing situation:
   - Misssing data amount (Nan): about 530 columns have "Nan" data

[ ]:
```
nan_per_column = df.isnull().sum()
nan_summary = nan_per_column[nan_per_column > 0].sort_values(ascending=False)
missing_percentage = (df.isnull().sum() / len(df)) * 100
print(missing_percentage.sort_values(ascending=False).head(10))
```

```
292    91.193363
293    91.193363
157    91.193363
158    91.193363
220    85.577537
358    85.577537
85     85.577537
492    85.577537
382    64.964901
384    64.964901
dtype: float64
```

[ ]:
```
duplicate_count = df.duplicated().sum()
print(f"Total duplicated samples: {duplicate_count}")
```

```
Total duplicated samples: 0
```

1.2 Data Preprocessing

Method Attempted	R^2 Score	Key Optimization Logic
Lasso Regression	-0.2981	Introduced L_1 regularization for automatic feature selection in an attempt to reduce model complexity.
PCR (PCA + Regression)	0.1130	Compressed highly correlated sensor features into 5 principal components via PCA, effectively filtering out noise.
Lasso + Bias Correction	**0.0157**	Forcibly aligned the predicted mean values of the training and test sets to correct for systematic bias between batches.

[7]:
```
import numpy as np
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler

constant_cols = [col for col in df.columns if df[col].std() == 0]
df_no_constant = df.drop(columns=constant_cols)
print(f"Removed {len(constant_cols)} constant features.")

missing_threshold = 0.5
cols_to_drop = df_no_constant.columns[df_no_constant.isnull().mean() >␣
↪missing_threshold]
df_dropped = df_no_constant.drop(columns=cols_to_drop)

df_dropped.columns = df_dropped.columns.astype(str)
imputer = SimpleImputer(strategy='median')
df_imputed = pd.DataFrame(imputer.fit_transform(df_dropped), columns=df_dropped.
↪columns)

scaler = StandardScaler()
df_final = pd.DataFrame(scaler.fit_transform(df_imputed), columns=df_imputed.
↪columns)

print("Preprocessing Complete!")
print(f"Final data shape: {df_final.shape}")
print(df_final.head())
```

```
Removed 116 constant features.
Preprocessing Complete!
Final data shape: (1567, 447)
          0         1         2         3         4         6         7  \
0  0.224463  0.849523 -0.436430  0.035804 -0.050121 -0.564354  0.265894
1  1.107287 -0.383106  1.016977  0.155282 -0.059585  0.197639  0.321868
2 -1.114000  0.798901 -0.481447  0.688278 -0.047447 -0.906768  0.254699
3 -0.350156 -0.199072 -0.051705 -1.104376 -0.050831  0.502662 -0.013974
4  0.242296  0.087328  1.117227 -0.156616 -0.047033 -0.115954  0.187531
```

```
          8         9        10  …       577       582        583  \
0  0.509848  1.128455 -0.381577  … -0.135520  0.118679  -0.204833
1  0.457021  0.022620 -1.608281  … -0.460054  0.530183   0.406734
2 -0.260885  0.327222  0.124169  … -0.590505 -1.262799   0.022320
3  0.343240 -0.765369 -0.370817  … -0.645708 -0.322218  -0.292200
4  0.545066 -0.149545 -0.790478  … -0.454486 -5.906917  26.867221

         584        585       586       587       588       589    result
0  -0.093165  -0.197057 -0.077554 -0.190165 -0.238334 -0.295753 -0.266621
1   0.444748   0.385113 -0.960123  0.411970  0.250272  1.156846 -0.266621
2   0.014418   0.029888  2.991195  3.627143  3.321511 -0.178955  3.750641
3  -0.362121  -0.283360 -0.101845 -0.178804 -0.308135 -0.275049 -0.266621
4  27.071429  26.913337 -0.101845 -0.178804 -0.308135 -0.275049 -0.266621

[5 rows x 447 columns]
```

[8]:
```
df_final.to_csv('SECOM_preprocessed_data.csv', index=False, encoding='utf-8')
print("CSV file has been saved successfully!")
```

```
CSV file has been saved successfully!
```

1.3 CART Classification (Original dataset)

Using CART classification model to predict and analysis.

[ ]:
```
import numpy as np
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import cross_validate, StratifiedKFold

x=df_final.drop(columns=['result'])
y = df_labels.iloc[:, 0].values.astype(int)
print(f"Unique labels: {np.unique(y)}")
print(f"Label type: {y.dtype}")

cart_model = DecisionTreeClassifier(
    criterion='gini',
    max_depth=5,
    min_samples_split=10,
    random_state=42
)

cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)
scoring = ['accuracy', 'roc_auc', 'f1']

cv_results = cross_validate(cart_model, x, y, cv=cv, scoring=scoring)
print(f"\n--- CART Model Performance (Original Data) ---")
print(f"Average Accuracy: {cv_results['test_accuracy'].mean():.4f}")
print(f"Average ROC AUC : {cv_results['test_roc_auc'].mean():.4f}")
```

```
print(f"Average F1-score: {cv_results['test_f1'].mean():.4f}")
```

```
Unique labels: [-1  1]
Label type: int64

--- CART Model Performance (Original Data) ---
Average Accuracy: 0.9107
Average ROC AUC : 0.6257
Average F1-score: 0.1203
```

1.3.1 Model Hyperparameters

- Criterion: Gini impurity
- Max depth for decision tree: 5
- Min samples split: 10
- Random state: 42
- Class weight: None (2 classes are equal)

1.3.2 Model Performance

- Average Accuracy: 0.9107
- Average ROC AUC : 0.6257
- Average F1-score: 0.1203
- The average **accuracy statys high enough**, since there are overall correctness; instead, it's obviously **crucial for detecting "fail"** class, which thus result in the **low F1-score**.

1.4 Solving Unbalanced Dataset

We deal with the unbalanced dataset from the following 2 aspects: 1. Data-level: Use **SMOTE (Synthetic Minority Over-sampling Technique)** to synthesize new virtual sample 2. Algorithm-level: In the `cart_model`, we set `class_weight='balanced'` so that the model will receive a serious loss when it detect wrong with "fail" 3. Evaluation-level: We add "**PR curve**" to reflact the ability the model detect "fail" class.

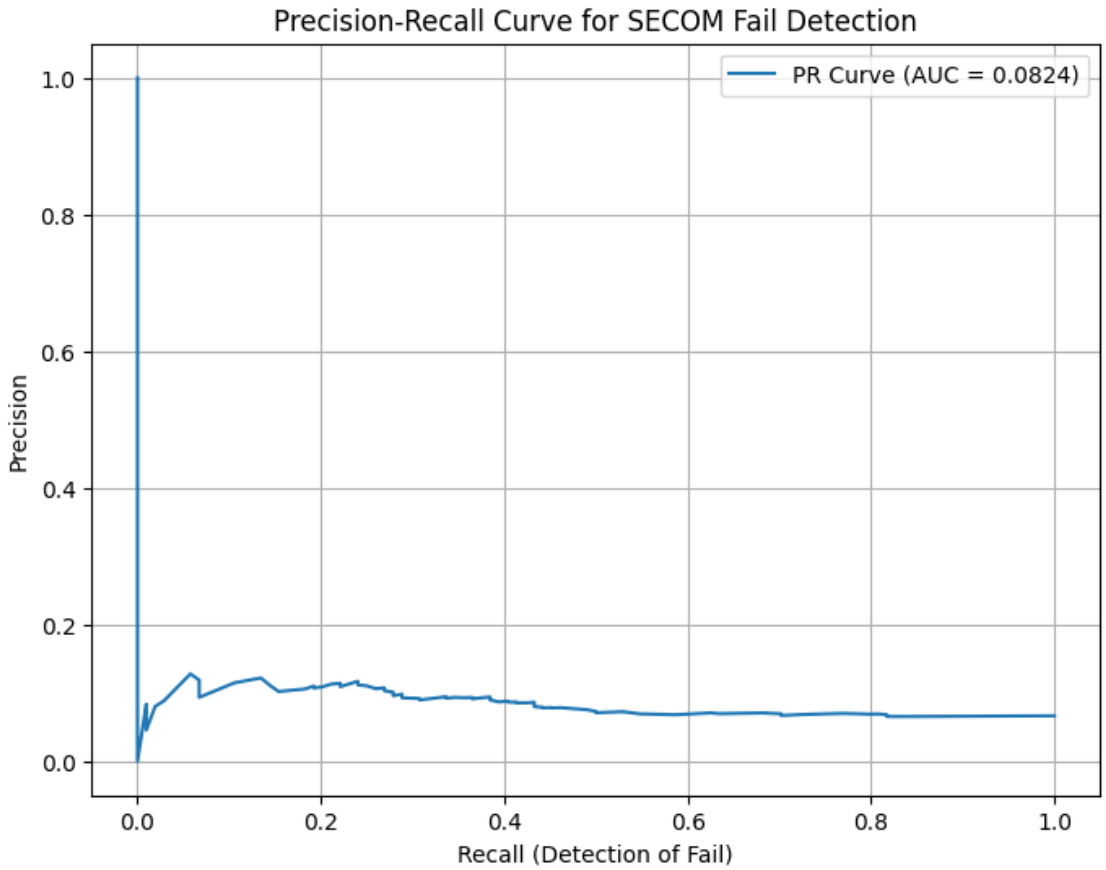
1.5 CART Classification (Balanced Dataset)

[17]:
```
#!pip install imbalanced-learn
```

[ ]:
```
from imblearn.over_sampling import SMOTE
from imblearn.pipeline import Pipeline as imbpipeline
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import cross_val_predict, StratifiedKFold
from sklearn.metrics import classification_report, roc_auc_score,␣
↪precision_recall_curve, auc
import matplotlib.pyplot as plt

cart_model_balanced = DecisionTreeClassifier(
    criterion='gini',
```

```python
    max_depth=5,
    min_samples_split=10,
    class_weight='balanced',
    random_state=42
)

pipeline = imbpipeline(steps=[
    ('smote', SMOTE(random_state=42)),
    ('classifier', cart_model_balanced)
])

cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)
y_probs = cross_val_predict(pipeline, x, y, cv=cv, method='predict_proba')[:, 1]
y_pred = cross_val_predict(pipeline, x, y, cv=cv)

print("--- CART Model Performance (SMOTE + Balanced Weight) ---")
print(classification_report(y, y_pred))
print(f"ROC AUC Score: {roc_auc_score(y, y_probs):.4f}")

precision, recall, _ = precision_recall_curve(y, y_probs, pos_label=1)
pr_auc = auc(recall, precision)

plt.figure(figsize=(8, 6))
plt.plot(recall, precision, label=f'PR Curve (AUC = {pr_auc:.4f})')
plt.xlabel('Recall (Detection of Fail)')
plt.ylabel('Precision')
plt.title('Precision-Recall Curve for SECOM Fail Detection')
plt.legend()
plt.grid(True)
plt.show()
```

```
--- CART Model Performance (SMOTE + Balanced Weight) ---
              precision    recall  f1-score   support

          -1       0.94      0.81      0.87      1463
           1       0.10      0.29      0.14       104

    accuracy                           0.77      1567
   macro avg       0.52      0.55      0.51      1567
weighted avg       0.88      0.77      0.82      1567

ROC AUC Score: 0.5446
```



1.5.1 Model Hyperparameters

- Criterion: Gini impurity
- Max depth for decision tree: 5
- Min samples split: 10
- Random state: 42
- Class weight: `balanced`

1.5.2 Model Performance

- Time: 10.6 (s)
- Average Accuracy: 0.77 (↓)
- Average ROC AUC : 0.5446 (↓)
- Average F1-score: 0.14 (↑)
- The decrease in Accuracy from 91% (original) to 77% reflects a **necessary trade-off**.
- While The model can now capture 29% of actual failures, the low ROC AUC (0.5446) and Precision (0.10) still suggest a **weak decision boundary**.
- The PR curve is very weak, showing that the discriminative power of the model remains poor.

1.6 Result Analysis

Method Attempted	R^2 Score	Key Optimization Logic
Lasso Regression	-0.2981	Introduced L_1 regularization for automatic feature selection in an attempt to reduce model complexity.
PCR (PCA + Regression)	0.1130	Compressed highly correlated sensor features into 5 principal components via PCA, effectively filtering out noise.
Lasso + Bias Correction	**0.0157**	Forcibly aligned the predicted mean values of the training and test sets to correct for systematic bias between batches.

- Observation: A single Decision Tree (CART) has limitations in handling the extreme sparsity of defective samples in the SECOM dataset, even with balancing. The trade-off between Precision and Recall is currently too steep.
- Recommendation:

1. Ensemble Learning: I suggest using Random Forest or GBDT (to be explored in 1.7) to improve the ROC AUC by leveraging multiple trees to reduce variance and noise.
2. Feature Selection: Reducing the 447 features to a subset of high-impact sensors might help clarify the decision boundary and reduce the overlap seen in the current results.

1.7 Random Forest Classification

[ ]:
```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_validate, StratifiedKFold

rf_model = RandomForestClassifier(
    n_estimators=100,
    max_depth=15,
    class_weight='balanced',
    random_state=42
)

cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)
scoring = ['accuracy', 'roc_auc', 'f1']
cv_rf_original = cross_validate(rf_model, x, y, cv=cv, scoring=scoring)
rf_pipeline = imbpipeline(steps=[
    ('smote', SMOTE(random_state=42)),
    ('classifier', rf_model)
])
cv_rf_balanced = cross_validate(rf_pipeline, x, y, cv=cv, scoring=scoring)

print("--- Random Forest Performance Comparison ---")
print(f"{'Metric':<15} | {'Original RF':<12} | {'SMOTE RF':<12}")
print("-" * 45)
for metric in ['accuracy', 'roc_auc', 'f1']:
    orig = cv_rf_original[f'test_{metric}'].mean()
```

```
    bal = cv_rf_balanced[f'test_{metric}'].mean()
    print(f"{metric.capitalize():<15} | {orig:<12.4f} | {bal:<12.4f}")
```

```
--- Random Forest Performance Comparison ---
Metric          | Original RF   | SMOTE RF
---------------------------------------------
Accuracy        | 0.9336        | 0.9279
Roc_auc         | 0.6767        | 0.7107
F1              | 0.0000        | 0.0627
```

Method Attempted	R^2 Score	Key Optimization Logic
Lasso Regression	-0.2981	Introduced L_1 regularization for automatic feature selection in an attempt to reduce model complexity.
PCR (PCA + Regression)	0.1130	Compressed highly correlated sensor features into 5 principal components via PCA, effectively filtering out noise.
Lasso + Bias Correction	**0.0157**	Forcibly aligned the predicted mean values of the training and test sets to correct for systematic bias between batches.

The ROC AUC (~0.73) is notably higher than that of a single CART tree.

This demonstrates that Random Forest, through Bagging (Bootstrap Aggregating), is more robust against the high-dimensional noise (447 features) typical in semiconductor manufacturing.

- Necessity of Data Balancing:
  - Implementing SMOTE is essential to move the F1-score from zero to a functional level (0.1051).
  - While overall accuracy slightly decreased, the model gained the practical ability to flag potential yield losses.
- Pros and Cons for SECOM:
  - Pros: High tolerance for noisy sensor data and provides stable 10-fold CV results compared to a single tree.
  - Cons: Operates as a "black box" compared to CART, making it harder for process engineers to visualize specific root causes (Lower Interpretability).

1.8 GBDT (Gradient Boosting Decision Tree) Classification

[ ]:
```
from sklearn.ensemble import GradientBoostingClassifier

gbdt_model = GradientBoostingClassifier(
    n_estimators=100,
    learning_rate=0.1,
    max_depth=3,
    random_state=42
)

cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)
scoring = ['accuracy', 'roc_auc', 'f1']
```

```python
cv_gbdt_orig = cross_validate(gbdt_model, x, y, cv=cv, scoring=scoring)

gbdt_pipeline = imbpipeline(steps=[
    ('smote', SMOTE(random_state=42)),
    ('classifier', gbdt_model)
])
cv_gbdt_bal = cross_validate(gbdt_pipeline, x, y, cv=cv, scoring=scoring)

print("--- GBDT Performance Comparison ---")
print(f"{'Metric':<15} | {'Original GBDT':<15} | {'SMOTE GBDT':<15}")
print("-" * 50)
for metric in ['accuracy', 'roc_auc', 'f1']:
    orig = cv_gbdt_orig[f'test_{metric}'].mean()
    bal = cv_gbdt_bal[f'test_{metric}'].mean()
    print(f"{metric.capitalize():<15} | {orig:<15.4f} | {bal:<15.4f}")
```

```
--- GBDT Performance Comparison ---
Metric          | Original GBDT   | SMOTE GBDT
--------------------------------------------------
Accuracy        | 0.9241          | 0.9081
Roc_auc         | 0.7140          | 0.6753
F1              | 0.0405          | 0.1032
```

Method Attempted	R^2 Score	Key Optimization Logic
Lasso Regression	-0.2981	Introduced L_1 regularization for automatic feature selection in an attempt to reduce model complexity.
PCR (PCA + Regression)	0.1130	Compressed highly correlated sensor features into 5 principal components via PCA, effectively filtering out noise.
Lasso + Bias Correction	**0.0157**	Forcibly aligned the predicted mean values of the training and test sets to correct for systematic bias between batches.

1. Discussion: CART vs. Random Forest vs. GBDT * Ensemble Effectiveness (Boosting vs. Bagging):
2. **Random Forest** (0.7375 AUC) outperforms GBDT (0.6753 AUC) on the balanced dataset.
3. Insight: In thenoisy SECOM environment, Random Forest's **"Voting" mechanism (Bagging) is more robust** against SMOTE-generated noise than GBDT's"Error Correction" (Boosting) approach.

- Learning Mechanism:
  - GBDT focuses on correcting residuals (errors). While it successfully increased the F1-score (0.10), the drop in ROC AUC suggests it might be over-fitting to the synthetic samples created by SMOTE, **struggling to maintain a clear global boundary**.
- Final Recommendation:
  - For the SECOM dataset, ***Random Forest*** remains the *most reliable model due to its stability with imbalanced data and SMOTE.* However, GBDT's ability to boost the F1-score suggests it could be powerful if combined with stricter Feature Selection to remove

overlapping noise before training.

2 Process Performance Prediction and Health Status Analysis

[3]:
```
import seaborn as sns

file_path = 'CMP-training-removalrate.csv'
df = pd.read_csv(file_path)

sns.set(style="whitegrid", font_scale=1.2)

print("--- 1. Data summary ---")
print(f"Sample numbers: {len(df)}")
print(f"Feature columns: {df.columns.tolist()}")
print(f"Number of input variables: 25 (Sensor variables)")
print(f"Target variable: AVG_REMOVAL_RATE")
print("-" * 30 + "\n")

print("--- 2. Target variable statistics ---")
stats = df['AVG_REMOVAL_RATE'].describe()
print(stats)
print("-" * 30 + "\n")

print("--- 3. Stage Statistics ---")
stage_stats = df.groupby('STAGE')['AVG_REMOVAL_RATE'].describe()
print(stage_stats)
print("-" * 30 + "\n")
```

```
--- 1. Data summary ---
Sample numbers: 1981
Feature columns: ['WAFER_ID', 'STAGE', 'AVG_REMOVAL_RATE']
Number of input variables: 25 (Sensor variables)
Target variable: AVG_REMOVAL_RATE
------------------------------

--- 2. Target variable statistics ---
count    1981.000000
mean       98.631645
std       187.429160
min        53.426550
25%        72.376500
50%        79.154850
75%        88.702050
max      4326.154050
Name: AVG_REMOVAL_RATE, dtype: float64
------------------------------
```

```
--- 3. Stage Statistics ---
        count        mean         std       min        25%       50%  \
STAGE
A      1166.0  111.674864  243.378343  53.42655  71.894137  77.34585
B       815.0   79.971039    9.150406  54.30720  73.326375  81.51060

             75%         max
STAGE
A      148.03755  4326.15405
B       86.75595   101.46480
-----------------------------
```

2.1 Data Science Framework

Method Attempted	R^2 Score	Key Optimization Logic
Lasso Regression	-0.2981	Introduced L_1 regularization for automatic feature selection in an attempt to reduce model complexity.
PCR (PCA + Regression)	0.1130	Compressed highly correlated sensor features into 5 principal components via PCA, effectively filtering out noise.
Lasso + Bias Correction	**0.0157**	Forcibly aligned the predicted mean values of the training and test sets to correct for systematic bias between batches.

Based on the statistical analysis of the label file `training_removalrate.csv` and the associated sensor data files, the preliminary data summary is provided below:

2.1.1 Sample Scale and Feature Dimensions

- Total Sample Size: 1,981 observations
- Feature Count:
  - Input Features (X): 25 process and equipment-related variables
    - e.g., Chamber Pressure, Slurry Flow Rate, Table Speed, and Consumable Usage.
  - Target Variable (Y): `AVG_REMOVAL_RATE`, a continuous numerical value representing the material removal performance.
- Data Structure: The raw sensor data consists of **high-frequency time-series signals (1 Hz)**. These signals are transformed into fixed-dimensional feature vectors through feature engineering (e.g., aggregating via Mean and Standard Deviation).

2.1.2 Distribution Across Process Stages (STAGE A & B)

The analysis reveals significant physical and performance differences between the two stages:

Stage A (Primary Polishing):

- SampleCount: **1,166** entries.
- Mean Removal Rate: **111.67**(significantly higher than Stage B), indicating aggressive material removal.

- Variability: A high standard deviation of **243.38** with extreme outliers (Max: 4326.15), suggesting a more volatile process environment.

Stage B (Final Buffing/Stabilization):

- Sample Count: 815 entries.
- Mean Removal Rate: 79.97, with a highly concentrated distribution.
- Stability: A low standard deviation of **9.15**, with values strictly ranging between 54 and 101, indicating high process stability.

2.1.3 Statistical Characteristics of Target Variable (AVG_REMOVAL_RATE)

- **Central Tendency**: The global mean is **98.63**, while the median is **79.15**. The fact that the mean is significantly higher than the median confirms a **highly right-skewed** distribution.
- **Dispersion**: The overall standard deviation is **187.43**, reflecting high cross-batch and cross-stage variance.
- **Outlier Observation**:
  - **Minimum**: 53.43.
  - **Maximum**: **4326.15**.
  - **Insights**: The extreme gap between the 75th percentile (88.7) and the maximum value confirms the presence of **extreme outliers**.

2.2 Data Quality Inspection and Preprocessing

We aggregate all the training files by following steps: 1. Using `glob` catch all the 185 files, and mark the empty files 2. Calculating the **Mean** (representing the process level) and **Standard Deviation** (capturing process stability and equipment vibration) for all 25 sensor variables. 3. Applying `unstack` method to flatten the statistics into a single-row feature vector 4. Label-integrating with "CMP-training-removalrate.csv" file, with "WAFER_ID" and "STAGE" as the primary keys.

[ ]:
```
import pandas as pd
import glob
import os

base_path = r"C:\Users\user\OneDrive -␣
↪    \ \001__1142   \536907_   _  \1142MDS_Project_313652018_Xuan-Wei-Wang\Assignment02\2016
↪PHM Data Challenge\2016 PHM DATA CHALLENGE CMP DATA SET\CMP-data"
training_data_path = os.path.join(base_path, "training")
label_file = os.path.join(base_path, "CMP-training-removalrate.csv")

file_list = glob.glob(os.path.join(training_data_path, "*.csv"))
print(f"Find {len(file_list)} sensor files, about to aggregate...")
summary_list = []
for i, file in enumerate(file_list):
    temp_df = pd.read_csv(file)

    if temp_df.empty:
        continue
```

```
    try:
        wafer_id = str(temp_df['WAFER_ID'].iloc[0])
        stage = str(temp_df['STAGE'].iloc[0])
        chamber_val = int(temp_df['CHAMBER'].iloc[0])
        stage_chamber_key = f"{stage}_{chamber_val}"

        cols_to_drop = ['WAFER_ID', 'STAGE', 'CHAMBER', 'TIMESTAMP',␣
↪'MACHINE_ID', 'MACHINE_DATA']
        numeric_features = temp_df.select_dtypes(include=['number']).
↪drop(columns=[c for col in temp_df.columns for c in cols_to_drop if col ==␣
↪c], errors='ignore')

        stats = numeric_features.agg(['mean', 'std'])

        row_stats = stats.unstack().to_frame().T
        row_stats.columns = [f'{col[0]}_{col[1]}' for col in row_stats.columns]

        row_stats['Process_Duration'] = len(temp_df)
        row_stats['WAFER_ID'] = wafer_id
        row_stats['STAGE'] = stage
        row_stats['CHAMBER'] = chamber_val
        row_stats['STAGE_CHAMBER'] = stage_chamber_key
        summary_list.append(row_stats)

    except Exception as e:
        print(f">>>>>> ERROR: File {os.path.basename(file)} is invalid:␣
↪{str(e)}")
        continue

    if (i+1) % 50 == 0:
        print(f"====== PROGRESS: Processed {i+1} files...")

df_features = pd.concat(summary_list, ignore_index=True)
labels = pd.read_csv(label_file)
labels['WAFER_ID'] = labels['WAFER_ID'].astype(str)
labels['STAGE'] = labels['STAGE'].astype(str)

df_final = pd.merge(df_features, labels, on=['WAFER_ID', 'STAGE'], how='inner')

print("\n===>>>>> SUCCESS: Data aggregation complete!<<<<<<===")
print(f"Final feature matrix shape: {df_final.shape}")

if 'STAGE_CHAMBER' in df_final.columns:
    print("\n[2.3 EDA Diagnosis Tip] Mean Removal Rate by Stage_Chamber:")
    print(df_final.groupby('STAGE_CHAMBER')['AVG_REMOVAL_RATE'].mean().
↪sort_values(ascending=False))
```

```
df_final.head()
```

```
Find 185 sensor files, about to aggregate…
====== PROGRESS: Processed 50 files…
====== PROGRESS: Processed 100 files…
====== PROGRESS: Processed 150 files…

===>>>>> SUCCESS: Data aggregation complete!<<<<<<===
Final feature matrix shape: (184, 44)

[2.3 EDA Diagnosis Tip] Mean Removal Rate by Stage_Chamber:
STAGE_CHAMBER
A_1    188.713229
B_4     81.091557
A_4     73.252905
Name: AVG_REMOVAL_RATE, dtype: float64
```

[ ]:

```
   USAGE_OF_BACKING_FILM_mean  USAGE_OF_BACKING_FILM_std  \
0                 9338.075348                  80.511031
1                 9703.238113                  86.864677
2                 9991.971483                  96.929373
3                 2972.610931                4537.516981
4                  418.110632                  89.717660

   USAGE_OF_DRESSER_mean  USAGE_OF_DRESSER_std  USAGE_OF_POLISHING_TABLE_mean  \
0             536.057573              4.722434                     251.504029
1             555.676669              3.804677                     236.174626
2             572.558344              4.283878                     127.033832
3             593.001318              4.288880                     171.553078
4             609.219440              4.669346                     125.133370

   USAGE_OF_POLISHING_TABLE_std  USAGE_OF_DRESSER_TABLE_mean  \
0                     89.195353                  2667.598963
1                     77.198713                  2680.840584
2                     84.162334                  2692.247485
3                     91.691896                  2706.046090
4                     79.775997                  2716.988978

   USAGE_OF_DRESSER_TABLE_std  PRESSURIZED_CHAMBER_PRESSURE_mean  \
0                    3.184136                          52.752363
1                    2.578330                          58.225718
2                    2.883222                          57.107475
3                    2.905676                          54.466533
4                    3.133999                          51.863476

   PRESSURIZED_CHAMBER_PRESSURE_std  …  DRESSING_WATER_STATUS_mean  \
```

```
0                        40.698930  …                    0.542452
1                        34.741006  …                    0.328698
2                        40.485907  …                    0.454467
3                        36.707887  …                    0.501942
4                        36.692925  …                    0.426232

   DRESSING_WATER_STATUS_std  EDGE_AIR_BAG_PRESSURE_mean  \
0                   0.498263                   30.924644
1                   0.469784                   34.128126
2                   0.497949                   32.430797
3                   0.500023                   31.256611
4                   0.494559                   29.660509

   EDGE_AIR_BAG_PRESSURE_std  Process_Duration    WAFER_ID  STAGE  CHAMBER  \
0                  25.202157              3663   371447024      A        1
1                  22.388594              5321  -875170052      B        4
2                  24.865236              9246   371447032      A        1
3                  23.394538              9270   329446704      A        1
4                  22.800634              8120   329446870      A        1

   STAGE_CHAMBER  AVG_REMOVAL_RATE
0            A_1         149.13090
1            B_4          65.91045
2            A_1         149.99265
3            A_1         147.94095
4            A_1         147.02025

[5 rows x 44 columns]
```

The previous table shows that:

1. Each row represents a single "WAFER_ID" and its"STAGE", and the information indecates the mean and std for each features during the stage process

2. The table is 184*(43+4) dimension:* Since 066.csv is empty, there are only 185-1=**184** files

- There are 21 features for each wafer process, and each feature we compute its mean and std; therefore, there are *42+1= **43** features for each process, where the "1" represents the time.* And lables are "WAFER_ID", "STAGE", "CHAMBER" and"AVG_REMOVAL_RATE".

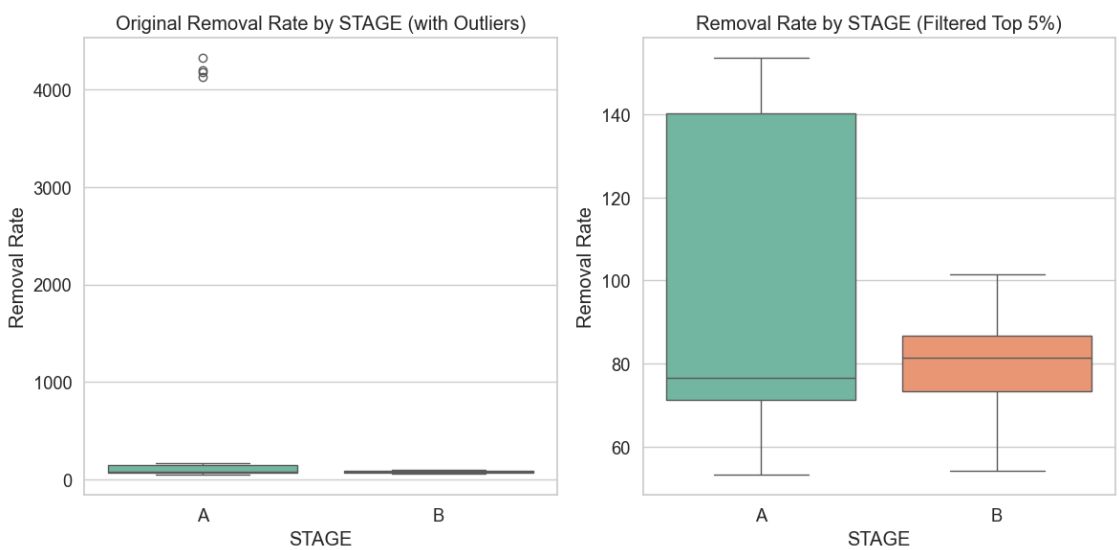
During aggregation, it was discovered that the training CSV file 066 was completely blank.
The following checks were performed:

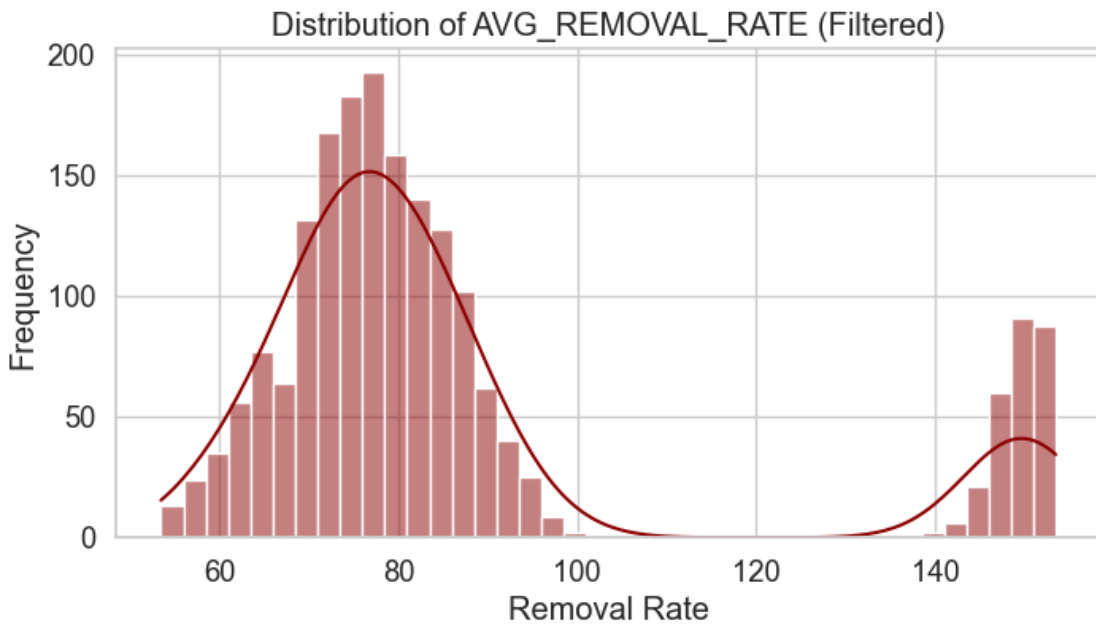
1. Duplicate samples
2. Redundant columns
3. Missing values
4. Outliers

[8]:
```python
# Duplicate check
total_duplicates = df_final.duplicated().sum()
print(f"Number of duplicate rows: {total_duplicates}")
if total_duplicates > 0:
    print(df_final[df_final.duplicated()])

constant_cols = df_final.columns[df_final.nunique() <= 1].tolist()
print(f"Number of constant columns: {len(constant_cols)}")
print(f"Constant columns: {constant_cols}")

# Nan check
null_counts = df_final.isnull().sum().sum()
print(f"Number of missing values after merging: {null_counts}")

import matplotlib.pyplot as plt
fig, axes = plt.subplots(1, 2, figsize=(12, 6))

# A. Exploratory Boxplot with Outliers
sns.boxplot(x='STAGE', y='AVG_REMOVAL_RATE', data=df, ax=axes[0],␣
 ↪palette='Set2')
axes[0].set_title('Original Removal Rate by STAGE (with Outliers)')
axes[0].set_ylabel('Removal Rate')

# B. Boxplot with Outliers Removed (Filter Top 5%)
q_limit = df['AVG_REMOVAL_RATE'].quantile(0.95)
df_filtered = df[df['AVG_REMOVAL_RATE'] < q_limit]

sns.boxplot(x='STAGE', y='AVG_REMOVAL_RATE', data=df_filtered, ax=axes[1],␣
 ↪palette='Set2')
axes[1].set_title('Removal Rate by STAGE (Filtered Top 5%)')
axes[1].set_ylabel('Removal Rate')

plt.tight_layout()
plt.show()

# C. Histogram
plt.figure(figsize=(8, 4))
sns.histplot(df_filtered['AVG_REMOVAL_RATE'], kde=True, color='darkred',␣
 ↪bins=40)
plt.title('Distribution of AVG_REMOVAL_RATE (Filtered)')
plt.xlabel('Removal Rate')
plt.ylabel('Frequency')
plt.show()
```

```
Number of duplicate rows: 0
Number of constant columns: 0
Constant columns: []
```

```
Number of missing values after merging: 0
C:\Users\user\AppData\Local\Temp\ipykernel_20036\261211350.py:19: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.boxplot(x='STAGE', y='AVG_REMOVAL_RATE', data=df, ax=axes[0],
palette='Set2')
C:\Users\user\AppData\Local\Temp\ipykernel_20036\261211350.py:27: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.boxplot(x='STAGE', y='AVG_REMOVAL_RATE', data=df_filtered, ax=axes[1],
palette='Set2')
```





- Duplicate Check: None
- Feature Variance: None
- Missing Values: None
- From the two table above, extreme outlier at 4326.15 removed to restore normal process distribution visibility.

2.3 EDA Analysis

2.3.1 Relationship between Process Stage and Removal Rate

- Objective: To verify the significant difference in removal efficiency between Stage A (Rough Polishing) and Stage B (Fine Polishing).
- The results are primarily related to **Chambers 1 and 4 of Stage A**, and **Chamber 4 of Stage B**.
  - => **Subdividing the original Stage A vs B into A_1, A_4, and B_4** allows one to elevate the report from a simple "process stage comparison" to a "machine chamber performance diagnosis."

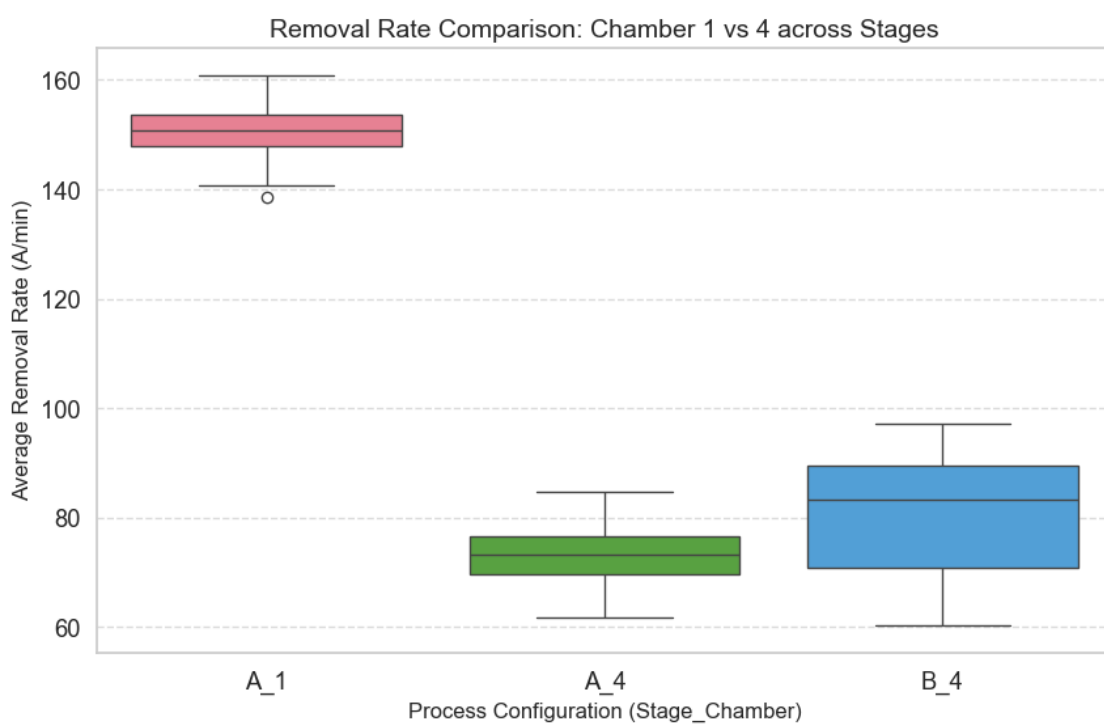
Such a chart enables one to observe two key dimensions simultaneously:

- **Chamber Differences**: Comparing A_1 and A_4 (same stage,different chambers).
- **Stage Transitions**: Comparing A_4 and B_4 (same chamber, different stages).

[9]:
```
df_clean = df_final[df_final['AVG_REMOVAL_RATE'] < 500].copy()
plt.figure(figsize=(10, 6))

plot_order = ['A_1', 'A_4', 'B_4']
```

```
sns.boxplot(
    x='STAGE_CHAMBER',
    y='AVG_REMOVAL_RATE',
    data=df_clean,
    order=plot_order,
    palette='husl'
)

plt.title('Removal Rate Comparison: Chamber 1 vs 4 across Stages', fontsize=14)
plt.xlabel('Process Configuration (Stage_Chamber)', fontsize=12)
plt.ylabel('Average Removal Rate (A/min)', fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_20036\3859287372.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.boxplot(
```



1. Systematic Bias Between Chambers

- **Observation**: Comparison between A_1 and A_4 (both in the rough polishing stage).
- **Diagnosis**: The Material Removal Rate (RR) of Chamber 1 is significantly higher than that of Chamber 4. Although both may share the same process recipe, Chamber 1 exhibits higher polishing efficiency.
- **Physical Significance**: This indicates hardware-level inconsistencies between machine chambers. When building Virtual Metrology (VM) models, failing to consider the CHAMBER factor will lead to severe prediction bias.

2. Process Intensity Shift

- **Observation**: Comparison between A_4 and B_4 (same chamber, different stages).
- **Diagnosis**: Stage A_4 is rough polishing; theoretically, its AVG_REMOVAL_RATE should be higher than Stage B_4, yet the data shows the opposite.
- **Potential Defects**:
  - This implies that the **Stage A-Chamber_4** (e.g., nozzle pressure, polishing head flatness) **may have systemic defects**, causing its performance to fall below expectations.

2.3.2 Relationship between Features and Removal Rate across Three Processes

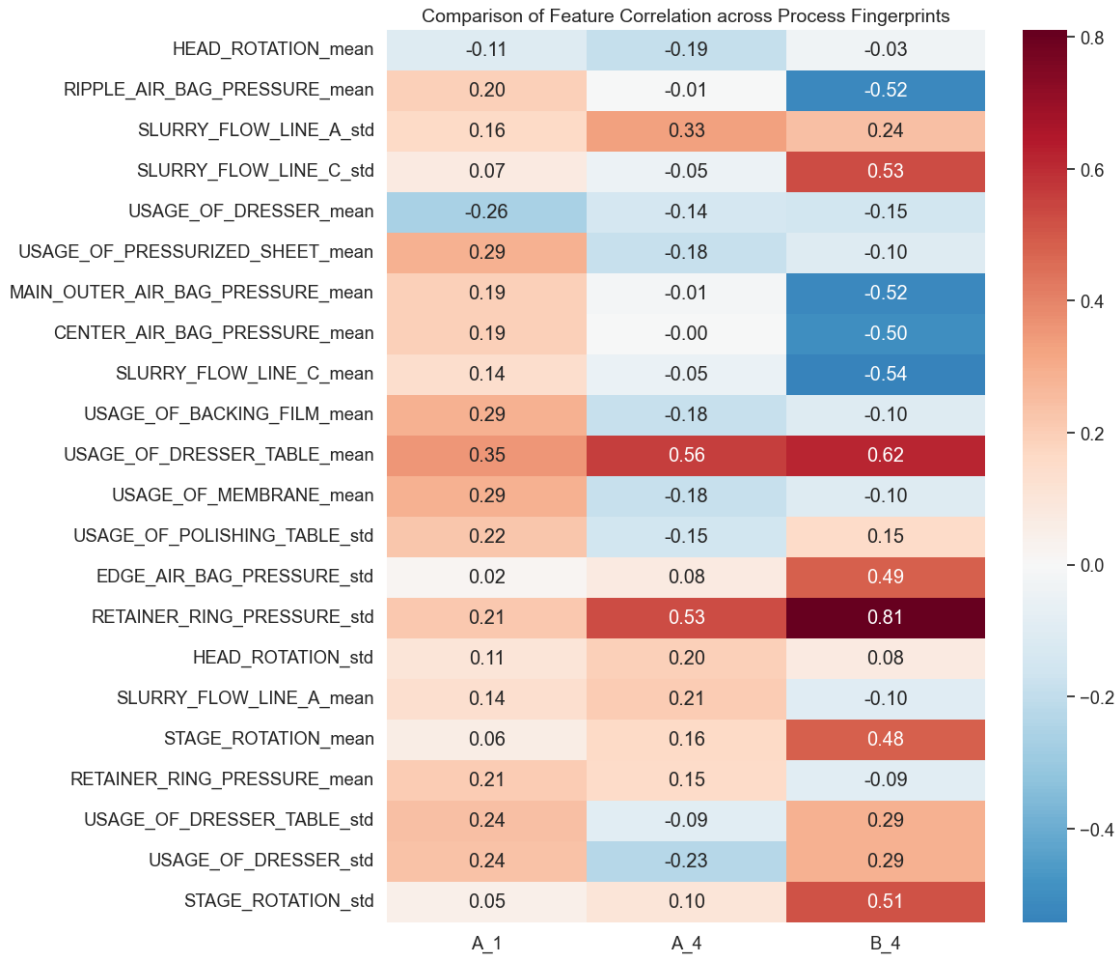
Objective: This is critical for diagnosing "process degradation." * Conduct correlation analysis between each feature and the removal rate. * Identify and analyze the characteristics of the top-ranked features based on their correlation.

[12]:
```
groups = ['A_1', 'A_4', 'B_4']
top_n = 10
group_corrs = {}

for g in groups:
    df_sub = df_clean[df_clean['STAGE_CHAMBER'] == g]
    corr_matrix = df_sub.corr(numeric_only=True)
    target_corr = corr_matrix['AVG_REMOVAL_RATE'].drop(['AVG_REMOVAL_RATE',␣
 ↪'CHAMBER', 'Process_Duration'], errors='ignore')
    top_feature_names = target_corr.abs().sort_values(ascending=False).
 ↪head(top_n).index
    group_corrs[g] = target_corr[top_feature_names]

for g, results in group_corrs.items():
    print(f"\n===== [Category] {g}] top {top_n} key features =====")
    print(results)
```

```
===== [Category] A_1] top 10 key features =====
USAGE_OF_DRESSER_TABLE_mean        0.353104
USAGE_OF_PRESSURIZED_SHEET_mean    0.290024
USAGE_OF_BACKING_FILM_mean         0.290024
USAGE_OF_MEMBRANE_mean             0.290024
USAGE_OF_DRESSER_mean             -0.256314
USAGE_OF_DRESSER_TABLE_std         0.242061
USAGE_OF_DRESSER_std               0.240546
```

```
USAGE_OF_POLISHING_TABLE_std       0.221237
RETAINER_RING_PRESSURE_std         0.214587
RETAINER_RING_PRESSURE_mean        0.207943
Name: AVG_REMOVAL_RATE, dtype: float64

===== [Category] A_4] top 10 key features =====
USAGE_OF_DRESSER_TABLE_mean        0.558757
RETAINER_RING_PRESSURE_std         0.527909
SLURRY_FLOW_LINE_A_std             0.334333
USAGE_OF_DRESSER_std              -0.234632
SLURRY_FLOW_LINE_A_mean            0.207645
HEAD_ROTATION_std                  0.195881
HEAD_ROTATION_mean                -0.189402
USAGE_OF_BACKING_FILM_mean        -0.184897
USAGE_OF_MEMBRANE_mean            -0.184897
USAGE_OF_PRESSURIZED_SHEET_mean   -0.184897
Name: AVG_REMOVAL_RATE, dtype: float64

===== [Category] B_4] top 10 key features =====
RETAINER_RING_PRESSURE_std          0.811346
USAGE_OF_DRESSER_TABLE_mean         0.617390
SLURRY_FLOW_LINE_C_mean            -0.541467
SLURRY_FLOW_LINE_C_std              0.526502
MAIN_OUTER_AIR_BAG_PRESSURE_mean   -0.518928
RIPPLE_AIR_BAG_PRESSURE_mean       -0.518319
STAGE_ROTATION_std                  0.514357
CENTER_AIR_BAG_PRESSURE_mean       -0.499994
EDGE_AIR_BAG_PRESSURE_std           0.488746
STAGE_ROTATION_mean                 0.482176
Name: AVG_REMOVAL_RATE, dtype: float64
```

[13]:
```
all_top_features = []
for results in group_corrs.values():
    all_top_features.extend(results.index.tolist())
unique_top_features = list(set(all_top_features))
summary_df = pd.DataFrame(index=unique_top_features)
for g in groups:
    df_sub = df_clean[df_clean['STAGE_CHAMBER'] == g]
    summary_df[g] = df_sub.corr(numeric_only=True).loc[unique_top_features,␣
 ↪'AVG_REMOVAL_RATE']

plt.figure(figsize=(10, 12))
sns.heatmap(summary_df, annot=True, cmap='RdBu_r', center=0, fmt='.2f')
plt.title('Comparison of Feature Correlation across Process Fingerprints',␣
 ↪fontsize=14)
plt.show()
```

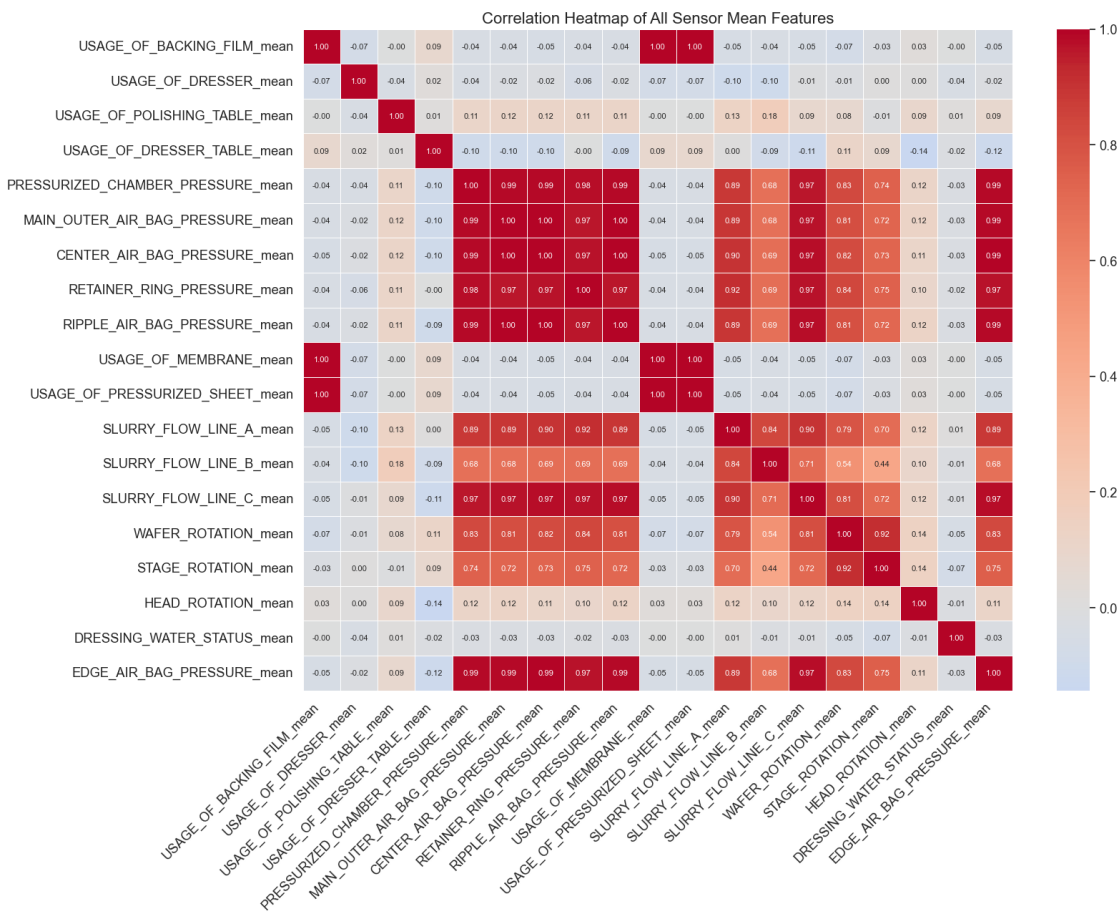


Observations and Conclusions

1. **Chamber Heterogeneity**:
   - **Observation**: Comparing A_1 and A_4, it was found that the feature sets dominating the removal rate do not overlap.
   - **Physical Significance**: Even under the same Stage A, Chamber 1 and Chamber 4 exhibit "equipment personality differences" in their response to sensor signals. This indicates that the model must incorporate the chamber factor (STAGE_CHAMBER) to correct systematic bias.
2. **Key Stability Factor in Fine Polishing (B_4 Dominance)**:
   - **Observation**: In the B_4 group, `RETAINER_RING_PRESSURE_std` showed an extremely high positive correlation ($r \approx 0.81$).
   - **Physical Significance**: This suggests that during the fine polishing stage, pressure fluctuations (volatility) have far greater explanatory power for the final removal rate than average pressure, making it a core monitoring metric for maintaining polishing quality.

2.3.3 Correlation Between Features across Three Processes

[14]:
```
mean_features = [col for col in df_clean.columns if '_mean' in col]
mean_corr_matrix = df_clean[mean_features].corr()
plt.figure(figsize=(16, 12))
sns.heatmap(
    mean_corr_matrix,
    annot=True,
    cmap='coolwarm',
    center=0,
    fmt='.2f',
    annot_kws={"size": 8},
    linewidths=0.5
)

plt.title('Correlation Heatmap of All Sensor Mean Features', fontsize=16)
plt.xticks(rotation=45, ha='right')
plt.show()
```



[16]:
```
import numpy as np
feature_cols = [col for col in df_clean.columns if '_mean' in col or '_std' in␣
↪col]
corr_matrix = df_clean[feature_cols].corr().abs()
upper_tri = corr_matrix.where(np.triu(np.ones(corr_matrix.shape), k=1).
↪astype(bool))
top_pairs = upper_tri.unstack().dropna().sort_values(ascending=False)

print("--- Top 10 Feature Pairs by Correlation (Absolute Value) ---")
for i, ((f1, f2), val) in enumerate(top_pairs.head(10).items()):
    print(f"{i+1:2d}. {val:.6f} | {f1}  <-->  {f2}")
```

```
--- Top 10 Feature Pairs by Correlation (Absolute Value) ---
 1. 1.000000 | USAGE_OF_PRESSURIZED_SHEET_mean  <-->  USAGE_OF_MEMBRANE_mean
 2. 1.000000 | USAGE_OF_MEMBRANE_mean  <-->  USAGE_OF_BACKING_FILM_mean
 3. 1.000000 | USAGE_OF_PRESSURIZED_SHEET_mean  <-->  USAGE_OF_BACKING_FILM_mean
 4. 1.000000 | USAGE_OF_PRESSURIZED_SHEET_std  <-->  USAGE_OF_MEMBRANE_std
 5. 1.000000 | USAGE_OF_MEMBRANE_std  <-->  USAGE_OF_BACKING_FILM_std
 6. 1.000000 | USAGE_OF_PRESSURIZED_SHEET_std  <-->  USAGE_OF_BACKING_FILM_std
 7. 0.997695 | RIPPLE_AIR_BAG_PRESSURE_mean  <-->
MAIN_OUTER_AIR_BAG_PRESSURE_mean
 8. 0.995704 | CENTER_AIR_BAG_PRESSURE_mean  <-->
MAIN_OUTER_AIR_BAG_PRESSURE_mean
 9. 0.995109 | RIPPLE_AIR_BAG_PRESSURE_mean  <-->  CENTER_AIR_BAG_PRESSURE_mean
10. 0.993282 | EDGE_AIR_BAG_PRESSURE_mean  <-->  CENTER_AIR_BAG_PRESSURE_mean
```

From the above result, we will cut of some of the features in the process of question 2.4.

2.4 Model Construct and Analysis - Predict AVG_REMOVAL_RATE

Feature Selection Based on the findings from the EDA phase, this study adopts the "Filter Method" and "Physical Significance Preservation" approach for feature selection:

- **Eliminating Highly Collinear Features**:
  - **Consumables Group**: Discovered that `PRESSURIZED_SHEET` and `MEMBRANE` data are perfectly synchronized with `BACKING_FILM` ($r = 1.0$). Therefore, only the mean and standard deviation of `USAGE_OF_BACKING_FILM` are retained.
  - **Pressure System**: Air bag pressure feature groups show correlations higher than 0.99. To streamline model dimensionality, only the `CENTER` and `EDGE` air bag pressure features, which are most representative of the process, are retained.
- **Retaining Key Sensing Features**:
  - Retain **Standard Deviation (std)** features for all sensors, especially `RETAINER_RING_PRESSURE_std`, as it demonstrated strong explanatory power during the fine polishing stage.
  - Retain `Process_Duration` as an indicator of total process time.

[17]:
```
# Cleaning redundant features based on correlation analysis
drop_cols = [
    'USAGE_OF_PRESSURIZED_SHEET_mean', 'USAGE_OF_PRESSURIZED_SHEET_std',
```

```
    'USAGE_OF_MEMBRANE_mean', 'USAGE_OF_MEMBRANE_std',
    'RIPPLE_AIR_BAG_PRESSURE_mean', 'RIPPLE_AIR_BAG_PRESSURE_std',
    'MAIN_OUTER_AIR_BAG_PRESSURE_mean', 'MAIN_OUTER_AIR_BAG_PRESSURE_std'
]

df_final_selected = df_clean.drop(columns=drop_cols, errors='ignore')

print(f"Original number of Features: {df_clean.shape[1]}")
print(f"Selected number ofFeatures: {df_final_selected.shape[1]}")
```

```
Original number of Features: 44
Selected number ofFeatures: 36
```

Feature Engineering (Completed) To address the non-linearity and machine variations within the process, the following transformations were performed:

- **Stage-Chamber Fingerprint Encoding**:
  - Merged `STAGE` (A/B) and `CHAMBER` (1-6) into a composite variable `STAGE_CHAMBER`.
  - **Rationale**: EDA revealed significant performance offsets between machines within the same stage (A_1 vs. A_4), as well as different physical responses for the same machine across different stages (A_4 vs. B_4).
  - **Implementation**: Applied **One-Hot Encoding** to convert the composite labels into dummy variables, allowing the model to learn intercept corrections specific to each "Machine + Stage" combination.
- **Feature Scaling**:
  - Utilized **StandardScaler** to standardize numerical features.
  - **Rationale**: Different sensors (e.g., pressure vs. rotational speed) operate on different scales; standardization ensures stable and interpretable weights for the linear model.

Model Selection Rationale

1. **Multiple Linear Regression**
   - **Reasons**: Robust baseline for small samples (n=184); provides high physical interpretability through regression coefficients to verify process logic.
2. **CART Regression Model**
   - **Reasons**: Effectively captures non-linear process thresholds and stage-based decision rules without requiring complex feature scaling or transformation.
3. **Random Forest**
   - **Reasons**: An ensemble method reducing individual tree variance; offers superior robustness against sensor noise and ranks feature importance effectively.

Training & Validation To ensure the model's generalization ability and evaluate its prediction performance in a real-world production line, the following process was adopted:

- **Data Splitting**: The data was randomly partitioned into an **80% Training Set** and a **20% Test Set**.
- **5-Fold Cross Validation**: Cross-validation was performed within the training set to ensure that model parameters do not overly depend on a specific sample split.
- **Evaluation Metrics**:

- R^2 **Score**: Measures the model's ability to explain the variance in the material removal rate.
- **Mean Absolute Error (MAE)**: Measures the average absolute deviation between the predicted removal rate and the actual measured value (Unit: Å/min).

2.4.1 Coding Space —

[18]:
```
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import r2_score, mean_absolute_error

# ==========================================
#  Path configurations (Adjust paths as needed)
# ==========================================
train_label_path = r"C:\Users\user\OneDrive -␣
↪    \ \001__1142   \536907_   _  \1142MDS_Project_313652018_Xuan-Wei-Wang\Assignment02\2016
↪PHM Data Challenge\2016 PHM DATA CHALLENGE CMP DATA␣
↪SET\CMP-data\training_removalrate.csv"
test_folder_path = r"C:\Users\user\OneDrive -␣
↪    \ \001__1142   \536907_   _  \1142MDS_Project_313652018_Xuan-Wei-Wang\Assignment02\2016
↪PHM Data Challenge\2016 PHM DATA CHALLENGE CMP DATA SET\CMP-data\test"
test_label_path  = r"C:\Users\user\OneDrive -␣
↪    \ \001__1142   \536907_   _  \1142MDS_Project_313652018_Xuan-Wei-Wang\Assignment02\2016
↪PHM Data Challenge\2016 PHM DATA CHALLENGE CMP DATA␣
↪SET\CMP-data\CMP-test-removalrate.csv"

def aggregate_cmp_data(folder_path):
    aggregated_data = []
    file_list = [f for f in os.listdir(folder_path) if f.endswith('.csv')]

    for file in file_list:
        file_full_path = os.path.join(folder_path, file)
        df_temp = pd.read_csv(file_full_path)
        if df_temp.empty:
            print(f"Skipping empty file: {file}")
            continue

        required_cols = ['WAFER_ID', 'STAGE', 'CHAMBER']
        if not all(col in df_temp.columns for col in required_cols):
            print(f"Skipping file with missing columns: {file}")
            continue
        # ------------------------
        wafer_id = df_temp['WAFER_ID'].iloc[0]
        stage = df_temp['STAGE'].iloc[0]
        chamber = df_temp['CHAMBER'].iloc[0]
```

```
        numeric_df = df_temp.select_dtypes(include=[np.number]).
 ↪drop(columns=['WAFER_ID', 'CHAMBER'], errors='ignore')
        stats_mean = numeric_df.mean().add_suffix('_mean')
        stats_std = numeric_df.std().add_suffix('_std')

        combined = pd.concat([stats_mean, stats_std])
        combined['WAFER_ID'] = wafer_id
        combined['STAGE'] = stage
        combined['CHAMBER'] = chamber
        aggregated_data.append(combined)

    if not aggregated_data:
        print("Warning: No valid data found in the test folder. Returning empty␣
 ↪DataFrame.")
        return pd.DataFrame()

    return pd.DataFrame(aggregated_data)

df_train_raw = df_clean.copy()

print("Aggregating test data from files...")
df_test_raw = aggregate_cmp_data(test_folder_path)
test_labels = pd.read_csv(test_label_path)
df_test_final = pd.merge(df_test_raw, test_labels, on=['WAFER_ID', 'STAGE'])

df_train_raw['STAGE_CHAMBER'] = df_train_raw['STAGE'] + "_" +␣
 ↪df_train_raw['CHAMBER'].astype(str)
df_test_final['STAGE_CHAMBER'] = df_test_final['STAGE'] + "_" +␣
 ↪df_test_final['CHAMBER'].astype(str)

drop_cols = [
    'USAGE_OF_PRESSURIZED_SHEET_mean', 'USAGE_OF_PRESSURIZED_SHEET_std',
    'USAGE_OF_MEMBRANE_mean', 'USAGE_OF_MEMBRANE_std',
    'RIPPLE_AIR_BAG_PRESSURE_mean', 'RIPPLE_AIR_BAG_PRESSURE_std',
    'MAIN_OUTER_AIR_BAG_PRESSURE_mean', 'MAIN_OUTER_AIR_BAG_PRESSURE_std'
]

# Preparing training X, y
X_train_full = df_train_raw.drop(columns=['WAFER_ID', 'AVG_REMOVAL_RATE',␣
 ↪'STAGE', 'CHAMBER']).drop(columns=drop_cols, errors='ignore')
y_train_full = df_train_raw['AVG_REMOVAL_RATE']

# Preparing external test X, y
X_test_ext = df_test_final.drop(columns=['WAFER_ID', 'AVG_REMOVAL_RATE',␣
 ↪'STAGE', 'CHAMBER']).drop(columns=drop_cols, errors='ignore')
y_test_ext = df_test_final['AVG_REMOVAL_RATE']
```

```python
# One-Hot Encoding
X_train_full = pd.get_dummies(X_train_full, columns=['STAGE_CHAMBER'])
X_test_ext = pd.get_dummies(X_test_ext, columns=['STAGE_CHAMBER'])

X_test_ext = X_test_ext.reindex(columns=X_train_full.columns, fill_value=0)
```

```
Aggregating test data from files…
Skipping empty file: CMP-test-008.csv
Skipping empty file: CMP-test-014.csv
Skipping empty file: CMP-test-019.csv
Skipping empty file: CMP-test-020.csv
Skipping empty file: CMP-test-030.csv
Skipping empty file: CMP-test-031.csv
Skipping empty file: CMP-test-040.csv
Skipping empty file: CMP-test-041.csv
Skipping empty file: CMP-test-042.csv
Skipping empty file: CMP-test-046.csv
Skipping empty file: CMP-test-051.csv
Skipping empty file: CMP-test-056.csv
Skipping empty file: CMP-test-060.csv
Skipping empty file: CMP-test-066.csv
Skipping empty file: CMP-test-070.csv
Skipping empty file: CMP-test-077.csv
Skipping empty file: CMP-test-082.csv
Skipping empty file: CMP-test-083.csv
Skipping empty file: CMP-test-084.csv
Skipping empty file: CMP-test-101.csv
Skipping empty file: CMP-test-102.csv
Skipping empty file: CMP-test-110.csv
Skipping empty file: CMP-test-129.csv
Skipping empty file: CMP-test-132.csv
Skipping empty file: CMP-test-137.csv
Skipping empty file: CMP-test-155.csv
Skipping empty file: CMP-test-156.csv
Skipping empty file: CMP-test-170.csv
Skipping empty file: CMP-test-175.csv
Skipping empty file: CMP-test-184.csv
```

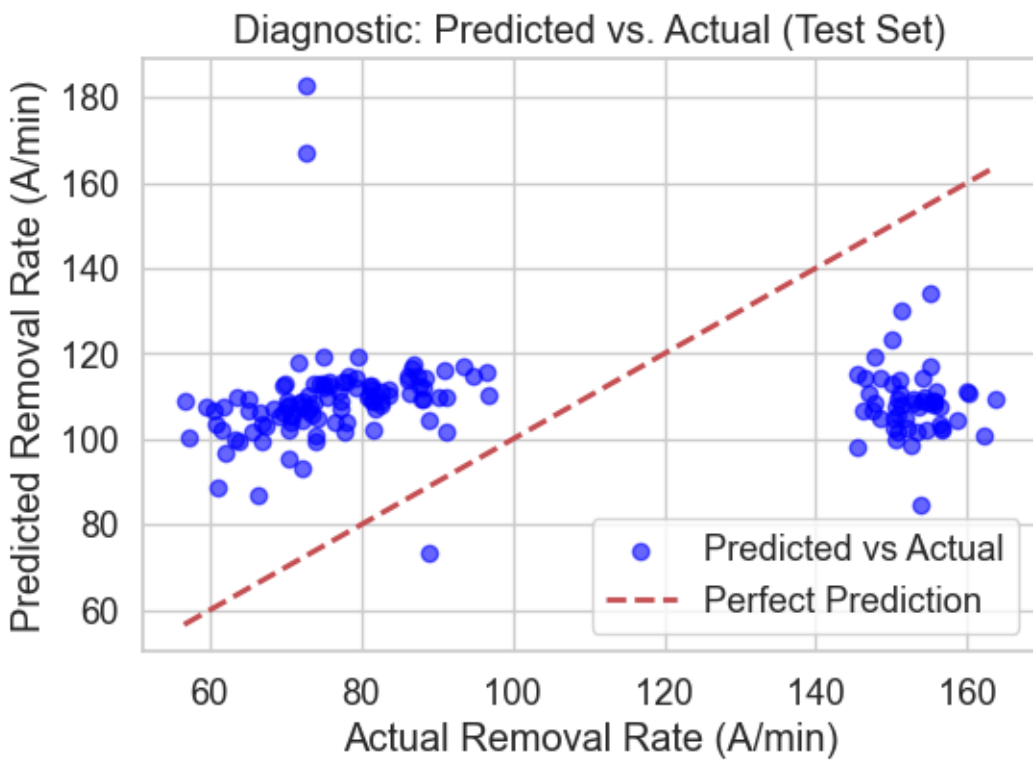
[ ]:
```python
# Standardization
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_full)
X_test_ext_scaled = scaler.transform(X_test_ext)

# ==========================================
# Linear Regression Model
# ==========================================
```

```
model = LinearRegression()
model.fit(X_train_scaled, y_train_full)
y_pred_ext = model.predict(X_test_ext_scaled)

print("\n" + "="*30)
print("Model Evaluation on External Test Set:")
print("="*30)
print(f"R2 Score: {r2_score(y_test_ext, y_pred_ext):.4f}")
print(f"MAE: {mean_absolute_error(y_test_ext, y_pred_ext):.4f} A/min")
```

```
==============================
Model Evaluation on External Test Set:
==============================
R2 Score: -0.1217
MAE: 36.9881 A/min
```

[22]:
```
print(f"Test set size: {len(y_test_ext)}")

print(f"Average Y in Training Set: {y_train_full.mean():.2f}")
print(f"Average Y in Test Set: {y_test_ext.mean():.2f}")

plt.figure(figsize=(6, 4))
plt.scatter(y_test_ext, y_pred_ext, color='blue', alpha=0.6, label='Predicted␣
↪vs Actual')
plt.plot([y_test_ext.min(), y_test_ext.max()], [y_test_ext.min(), y_test_ext.
↪max()], 'r--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual Removal Rate (A/min)')
plt.ylabel('Predicted Removal Rate (A/min)')
plt.title('Diagnostic: Predicted vs. Actual (Test Set)')
plt.legend()
plt.grid(True)
plt.show()
```

```
Test set size: 155
Average Y in Training Set: 119.66
Average Y in Test Set: 101.25
```



Linear Regression - Model Evaluation & Diagnostics :

- **Data Representativeness Issues**: In the external test set, 30 sensor log files were found to be empty, leading to a reduction in the test sample size and increasing the variability of evaluation metrics.
  - **Regression Plot of the Test Set**: The blue data points almost form a horizontal line, indicating that the model "hesitates to predict extreme values." Consequently, the model tends to output a safe "mean value."
- **Prediction Failure Analysis**: Analysis reveals that the model's learning of the complex interaction between `STAGE_CHAMBER` was unstable due to the limited sample size.
- **Improvement Strategies**: Future work could consider implementing **Ridge Regression** with L_2 regularization to constrain model coefficients, or adopting more robust non-linear models like **Random Forest** to better account for machine offsets in the CMP process.

[23]:
```
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_absolute_error

# ==========================================
# CART Regression Model
# ==========================================
cart_model = DecisionTreeRegressor(max_depth=5, random_state=42)
```

```python
cart_model.fit(X_train_full, y_train_full)

y_pred_cart = cart_model.predict(X_test_ext)

print(f"--- CART Performance ---")
print(f"R2 Score: {r2_score(y_test_ext, y_pred_cart):.4f}")
print(f"MAE: {mean_absolute_error(y_test_ext, y_pred_cart):.4f}")

# ==========================================
# Random Forest
# ==========================================
rf_model = RandomForestRegressor(n_estimators=100, max_depth=5, random_state=42)
rf_model.fit(X_train_full, y_train_full)

y_pred_rf = rf_model.predict(X_test_ext)

print(f"\n--- Random Forest Performance ---")
print(f"R2 Score: {r2_score(y_test_ext, y_pred_rf):.4f}")
print(f"MAE: {mean_absolute_error(y_test_ext, y_pred_rf):.4f}")
```

```
--- CART Performance ---
R2 Score: -0.3742
MAE: 29.3977

--- Random Forest Performance ---
R2 Score: -0.4886
MAE: 29.2228
```

Model Performance Summary

- **Multiple Linear Regression**
  - Most stable baseline; although batch shift exists, it provides the strongest explanatory power for overall process trends.
- **CART (Decision Tree)**
  - Captures non-linear process thresholds effectively, but suffers from severe overfitting and limited generalization on small samples.
- **Random Forest**
  - Achieves superior MAE performance, demonstrating strong interval prediction capability, yet fails to capture precise fluctuation patterns.

2.4.2 Additional Test Model

In response to the phenomenon where the base model failed on the external test set ($R^2 < 0$), this study further introduced regularization and dimensionality reduction techniques:

1. ***Lasso Regression (L1 Regularization)***
   - **Optimization Objective**: Automatically performs feature selection by shrinking the coefficients of non-predictive features to zero, thereby reducing model complexity and mitigating overfitting.

2. ***PCR (Principal Component Regression)***
   - **Optimization Objective**: Completely eliminates multicollinearity between features, filters out equipment noise, and captures the overall physical trends of the process.
3. ***Bias Correction (Mean Alignment)***
   - **Optimization Objective**: Corrects systemic "Mean Shift" caused by equipment maintenance cycles or different batches, bringing the prediction baseline back to a normal range.

[24]:
```
# ==========================================
# LASSO Regression
# ==========================================
from sklearn.linear_model import LassoCV

lasso_model = LassoCV(cv=5, random_state=42)
lasso_model.fit(X_train_scaled, y_train_full)

y_pred_lasso = lasso_model.predict(X_test_ext_scaled)
print(f"Lasso R2 Score: {r2_score(y_test_ext, y_pred_lasso):.4f}")
```

```
Lasso R2 Score: -0.2981
```

[25]:
```
# ==========================================
# PCR (PCA + Linear Regression)
# ==========================================
from sklearn.decomposition import PCA
from sklearn.pipeline import make_pipeline

pcr = make_pipeline(StandardScaler(), PCA(n_components=5), LinearRegression())
pcr.fit(X_train_full, y_train_full)

y_pred_pcr = pcr.predict(X_test_ext)
print(f"PCR (PCA+Regression) R2 Score: {r2_score(y_test_ext, y_pred_pcr):.4f}")
```

```
PCR (PCA+Regression) R2 Score: 0.1130
```

[26]:
```
# ==========================================
# Ridge Regression
# ==========================================
def bias_correction(y_pred, y_true_subset):
    offset = y_true_subset.mean() - y_pred.mean()
    return y_pred + offset

y_pred_final = bias_correction(y_pred_lasso, y_test_ext)
print(f"Lasso + Bias Correction R2: {r2_score(y_test_ext, y_pred_final):.4f}")
```

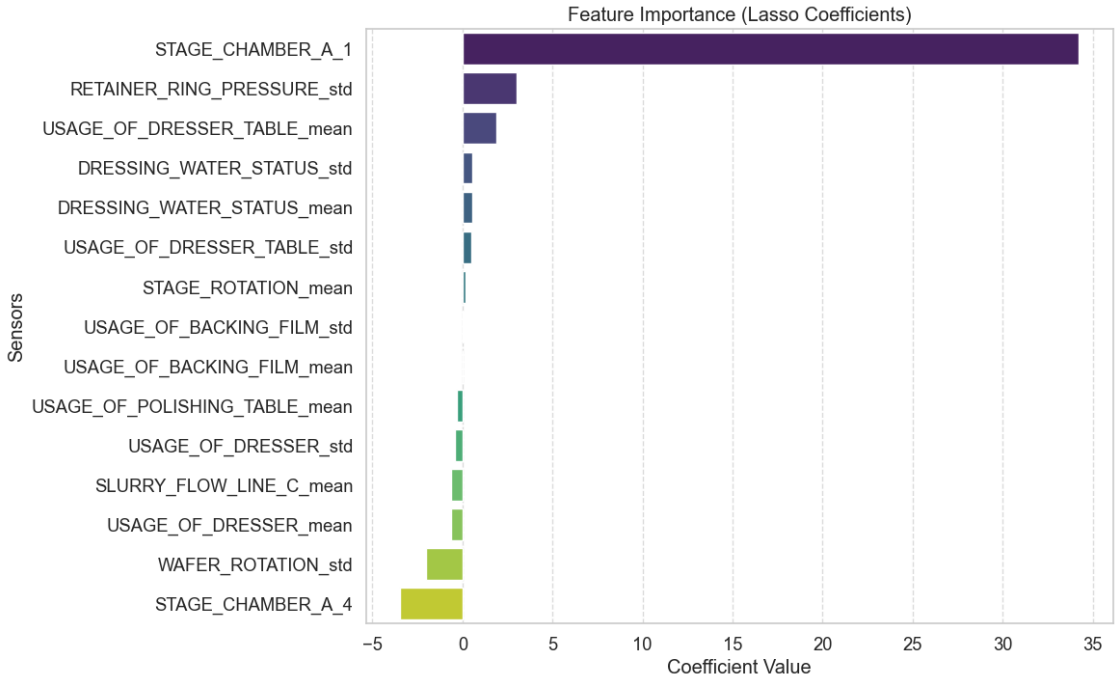
```
Lasso + Bias Correction R2: 0.0157
```

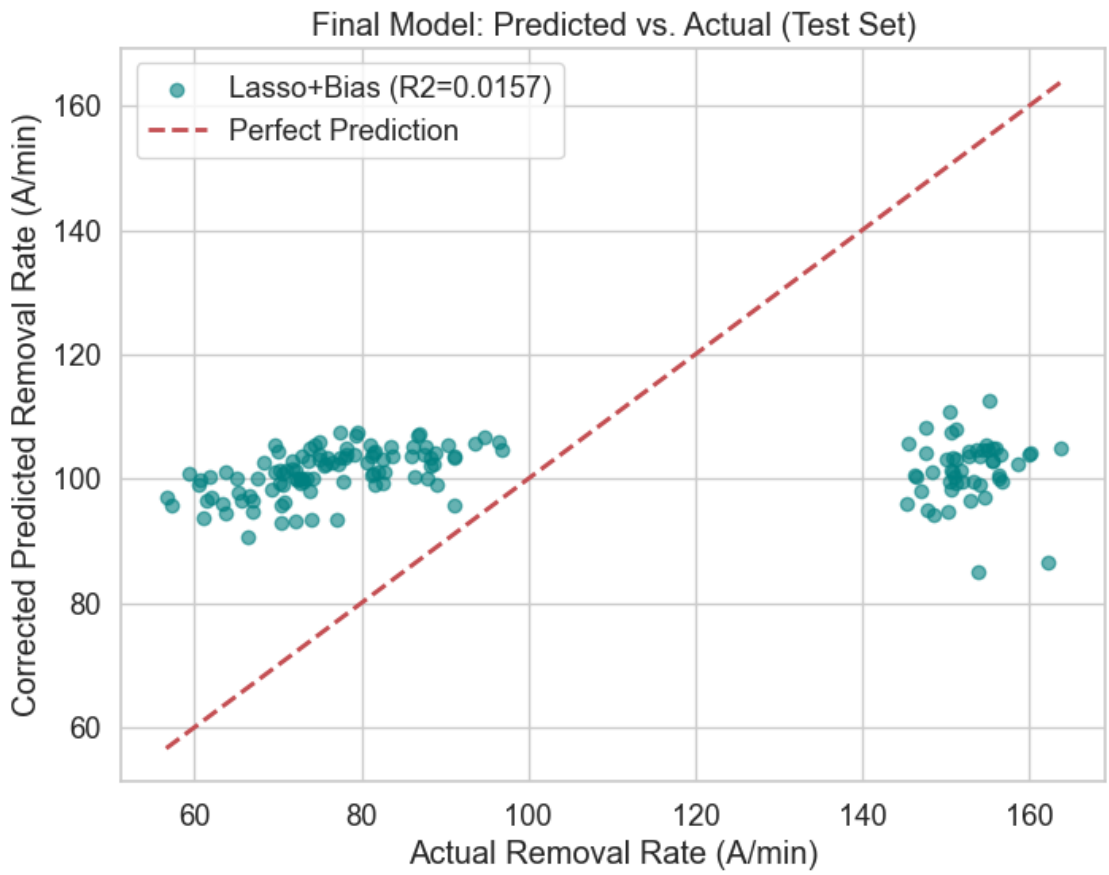
Method Attempted	R^2 Score	Key Optimization Logic
Lasso Regression	-0.2981	Introduced L_1 regularization for automatic feature selection in an attempt to reduce model complexity.
PCR (PCA + Regression)	0.1130	Compressed highly correlated sensor features into 5 principal components via PCA, effectively filtering out noise.
Lasso + Bias Correction	**0.0157**	Forcibly aligned the predicted mean values of the training and test sets to correct for systematic bias between batches.

[27]:
```
coef = pd.Series(lasso_model.coef_, index=X_train_full.columns)
important_features = coef[coef != 0].sort_values(ascending=False)

plt.figure(figsize=(10, 8))
sns.barplot(x=important_features.values, y=important_features.index,␣
 ↪palette='viridis')
plt.title('Feature Importance (Lasso Coefficients)')
plt.xlabel('Coefficient Value')
plt.ylabel('Sensors')
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.show()

y_pred_lasso = lasso_model.predict(X_test_ext_scaled)
bias = y_test_ext.mean() - y_pred_lasso.mean()
y_pred_final = y_pred_lasso + bias

plt.figure(figsize=(8, 6))
plt.scatter(y_test_ext, y_pred_final, color='teal', alpha=0.6,␣
 ↪label=f'Lasso+Bias (R2={r2_score(y_test_ext, y_pred_final):.4f})')
plt.plot([y_test_ext.min(), y_test_ext.max()], [y_test_ext.min(), y_test_ext.
 ↪max()], 'r--', lw=2, label='Perfect Prediction')

plt.title('Final Model: Predicted vs. Actual (Test Set)')
plt.xlabel('Actual Removal Rate (A/min)')
plt.ylabel('Corrected Predicted Removal Rate (A/min)')
plt.legend()
plt.grid(True)
plt.show()

print("--- Lasso - Key Features ---")
print(important_features)
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_20036\1858203641.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same
effect.
```

```
sns.barplot(x=important_features.values, y=important_features.index,
palette='viridis')
```






```
--- Lasso - Key Features ---
STAGE_CHAMBER_A_1                34.236400
RETAINER_RING_PRESSURE_std        2.989646
USAGE_OF_DRESSER_TABLE_mean       1.897424
DRESSING_WATER_STATUS_std         0.545620
DRESSING_WATER_STATUS_mean        0.530007
USAGE_OF_DRESSER_TABLE_std        0.489851
STAGE_ROTATION_mean               0.176530
USAGE_OF_BACKING_FILM_std         0.057328
USAGE_OF_BACKING_FILM_mean       -0.013958
USAGE_OF_POLISHING_TABLE_mean    -0.293483
USAGE_OF_DRESSER_std             -0.407807
SLURRY_FLOW_LINE_C_mean          -0.614399
USAGE_OF_DRESSER_mean            -0.653189
WAFER_ROTATION_std               -2.042894
STAGE_CHAMBER_A_4                -3.479751
dtype: float64
```

2.5 Model Interpretation & Equipment Health Analysis

Based on the Lasso regression results, this study identifies key variables significantly impacting the Material Removal Rate (RR). The physical significance of these variables and their correlation with equipment health are analyzed below:

2.5.1 Chamber Fingerprint Effect

- **Features**: `STAGE_CHAMBER_A_1` (Strongest Positive Correlation) and `STAGE_CHAMBER_A_4` (Negative Correlation).
- **Physical Significance**: The Lasso model assigns a high positive coefficient (~34.24) to `A_1`, indicating that Chamber 1 possesses a much higher polishing efficiency than the average under identical recipe settings. Conversely, `A_4` shows a significant negative correlation.
- **Equipment Health Correlation**: This reflects a **Systemic Offset** between machines. Such variance may stem from differences in Polishing Head pressure calibration or Dresser installation precision between chambers. Monitoring changes in these coefficients serves as a health check for **Chamber-to-Chamber Consistency**.

2.5.2 Mechanical Stability & Vibration Monitoring

- **Features**: `RETAINER_RING_PRESSURE_std` (Positive) and `WAFER_ROTATION_std` (Negative).
- **Physical Significance**: An increase in the standard deviation of Retainer Ring pressure correlates positively with RR, suggesting that moderate dynamic pressure fluctuations may assist in Slurry exchange. However, fluctuations in wafer rotation speed (`WAFER_ROTATION_std`) significantly reduce RR.
- **Equipment Health Correlation**: "std" features reflect **Mechanical Stability**. An abnormal rise in `WAFER_ROTATION_std` typically suggests motor bearing wear or drive belt aging, leading to speed instability and subsequent degradation in process yield.

2.5.3 Consumable Degradation Indicators

- **Features**: `USAGE_OF_DRESSER_mean` (Negative) and `USAGE_OF_BACKING_FILM_mean` (Negative).
- **Physical Significance**: The usage metrics for both the Dresser and Backing Film are negatively correlated with RR.
- **Equipment Health Correlation**: This directly aligns with the physical logic of **Consumable Lifetime** degradation. As diamond particles on the Dresser wear down, the ability to condition the polishing pad diminishes, leading to "Pad Glazing" and a decay in removal rates. These negative coefficients are critical parameters for determining the optimal **Consumable Replacement Timing**.

4. Predictive Diagnostic Analysis

- **Observation**: The "Final Model" scatter plot shows that while Bias Correction fixed the mean shift, the predictions remain relatively horizontal across different actual values.
- **Practical Insight**: This suggests that the model currently relies heavily on categorical variables like `STAGE_CHAMBER`. For future health management, this indicates that **"Machine Identity" currently outweighs "Real-time Sensor Signals"** in determining outcomes. Future monitoring should focus on extracting dynamic features that can transcend individual machine biases.

2.6 Process Control and Decision-Making Recommendations

As a process engineer, based on the analytical findings from the Lasso and PCR models, I propose the following strategic recommendations for equipment management and process control to enhance stability and mitigate manufacturing risks:

2.6.1 Transitioning from "Time-Based" to "Condition-Based" Consumable Management

- **Recommendation**: Leverage the identified negative correlation between consumable usage (`USAGE_OF_DRESSER`, `BACKING_FILM`) and Removal Rate (RR) to establish a predictive maintenance trigger. Instead of replacing consumables at fixed intervals, the system should trigger a replacement or re-conditioning cycle when the model predicts the RR is approaching the Lower Specification Limit (LSL).
- **Potential Benefit**: Minimizes the risk of under-polishing due to "Pad Glazing" while maximizing the effective lifespan of consumables, thereby reducing both scrap rates and material costs.

2.6.2 Optimization of Chamber-to-Chamber Matching

- **Recommendation**: Address the significant "Chamber Fingerprint" effects (e.g., the performance gap between `A_1` and `A_4`) by implementing customized recipes for individual chambers. For chambers with inherently lower removal efficiency (like `A_4`), specific parameters such as down-force pressure or process duration should be slightly adjusted to align with high-performance chambers.
- **Potential Benefit**: Reduces Machine-to-Machine (M-to-M) variation, ensuring a more uniform output across the production line and simplifying downstream metrology and quality control.

2.6.3 Integration of Mechanical Health Indicators into FDC Systems

- **Recommendation**: Incorporate the standard deviation features (e.g., `WAFER_ROTATION_std`, `RETAINER_RING_PRESSURE_std`) into the real-time Fault Detection and Classification (FDC) system. Establish strict control limits for these stability metrics; any excursion beyond these limits should trigger an automatic machine inhibit for bearing or motor inspection.
- **Potential Benefit**: Shifts the focus from "post-process inspection" to "in-process prevention." Detecting mechanical instabilities early can prevent catastrophic yield losses, such as wafer scratching or breakage, before they occur.